

## CAYO COCO, Cuba

WMO: 782390

Lat: 22.539N

Lon: 78.368W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (CUB)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-----|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |     |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p) |
| 1             | 15.7       | 17.5 | 9.4                           | 7.3 | 18.7 | 11.1 | 8.2 | 20.1 | 11.5                  | 22.3 | 10.5 | 22.4 | 1.9                   | 10   | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.0                    | 32.7            | 25.5 | 32.2 | 25.5 | 31.8 | 25.5 | 26.7                | 30.9 | 26.4 | 30.6 | 26.2 | 30.3 | 5.3                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.6                            | 20.8 | 28.9 | 25.2 | 20.4 | 28.8 | 25.0 | 20.1 | 28.7 | 83.2          | 31.2 | 82.0 | 30.7 | 80.9 | 30.4 | 28.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |      |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|------|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |      |  |
| (4) 10.1          | 9.2  | 8.3 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |      |  |
| (5)               |      |     | DB                         | 10.7 | 27.1               | 1.3 | 0.5                                                | 9.8 | 27.5       | 9.0 | 27.8       | 8.3 | 28.1       | 7.3 | 28.5 |  |
|                   |      |     | WB                         |      |                    |     |                                                    |     |            |     |            |     |            |     |      |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 26.1        | 23.1  | 23.8  | 24.2  | 25.9  | 26.6  | 28.0  | 28.7  | 28.7  | 28.1  | 27.0  | 25.0  | 24.2  |
|                                                                        | DBStd       | 2.49        | 2.14  | 1.90  | 1.82  | 1.36  | 1.35  | 1.27  | 1.03  | 1.01  | 0.96  | 1.27  | 1.70  | 2.21  |
|                                                                        | HDD10.0     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD18.3     | 3           | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     |
|                                                                        | CDD10.0     | 5878        | 406   | 386   | 439   | 477   | 513   | 539   | 578   | 581   | 542   | 526   | 452   | 439   |
|                                                                        | CDD18.3     | 2840        | 148   | 153   | 182   | 227   | 255   | 289   | 320   | 322   | 292   | 268   | 202   | 182   |
|                                                                        | CDH23.3     | 26209       | 673   | 837   | 1050  | 1832  | 2394  | 3252  | 3879  | 3897  | 3285  | 2602  | 1408  | 1101  |
|                                                                        | CDH26.7     | 7869        | 57    | 99    | 165   | 380   | 618   | 1177  | 1586  | 1609  | 1193  | 686   | 191   | 108   |
| (14) Wind                                                              | WSAvg       | 4.3         | 4.6   | 4.8   | 5.0   | 4.8   | 4.2   | 3.5   | 4.0   | 4.0   | 3.5   | 4.0   | 4.9   | 4.7   |
| (15) Precipitation                                                     | PrecAvg     | 932         | 30    | 31    | 30    | 48    | 102   | 136   | 102   | 121   | 137   | 108   | 52    | 36    |
|                                                                        | PrecMax     | 1359        | 62    | 84    | 98    | 126   | 300   | 328   | 188   | 173   | 193   | 236   | 165   | 116   |
|                                                                        | PrecMin     | 572         | 5     | 5     | 5     | 11    | 16    | 38    | 24    | 63    | 78    | 35    | 13    | 3     |
|                                                                        | PrecStd     | 167         | 16    | 21    | 21    | 25    | 65    | 64    | 36    | 27    | 31    | 54    | 35    | 26    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 29.1  | 29.8  | 31.2  | 31.5  | 32.0  | 33.3  | 33.2  | 33.2  | 32.7  | 31.7  | 30.2  | 29.4  |
|                                                                        |             | MCWB        | 22.4  | 21.9  | 22.7  | 23.2  | 24.7  | 25.4  | 25.6  | 25.6  | 25.8  | 25.2  | 25.2  | 23.9  |
|                                                                        | 2%          | DB          | 28.0  | 28.7  | 29.5  | 30.1  | 30.9  | 32.2  | 32.6  | 32.5  | 32.2  | 31.0  | 29.3  | 28.4  |
|                                                                        |             | MCWB        | 22.8  | 22.3  | 22.8  | 23.4  | 24.7  | 25.4  | 25.6  | 25.8  | 25.8  | 25.4  | 24.6  | 23.7  |
|                                                                        | 5%          | DB          | 27.2  | 27.9  | 28.5  | 29.5  | 30.1  | 31.6  | 32.1  | 32.1  | 31.6  | 30.4  | 28.5  | 27.8  |
|                                                                        |             | MCWB        | 22.5  | 22.5  | 22.5  | 23.3  | 24.3  | 25.4  | 25.5  | 25.8  | 25.7  | 25.2  | 24.0  | 23.5  |
|                                                                        | 10%         | DB          | 26.5  | 27.1  | 27.5  | 28.9  | 29.5  | 31.0  | 31.6  | 31.6  | 31.0  | 29.8  | 27.7  | 27.2  |
|                                                                        |             | MCWB        | 22.1  | 22.1  | 22.0  | 23.1  | 24.0  | 25.4  | 25.4  | 25.7  | 25.5  | 25.0  | 23.6  | 23.1  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 24.2  | 24.1  | 24.5  | 25.2  | 26.1  | 26.9  | 26.8  | 27.1  | 27.1  | 26.6  | 25.7  | 25.4  |
|                                                                        |             | MCDB        | 27.0  | 27.0  | 27.7  | 28.3  | 30.2  | 31.1  | 31.5  | 31.5  | 31.2  | 30.1  | 29.0  | 27.9  |
|                                                                        | 2%          | WB          | 23.6  | 23.6  | 23.9  | 24.6  | 25.5  | 26.4  | 26.4  | 26.6  | 26.6  | 26.2  | 25.3  | 24.6  |
|                                                                        |             | MCDB        | 26.5  | 26.7  | 27.4  | 28.3  | 29.2  | 30.5  | 31.1  | 31.0  | 30.6  | 29.6  | 28.3  | 27.0  |
|                                                                        | 5%          | WB          | 23.1  | 23.1  | 23.3  | 24.1  | 25.1  | 26.0  | 26.1  | 26.3  | 26.2  | 25.7  | 24.8  | 24.2  |
|                                                                        |             | MCDB        | 26.2  | 26.5  | 27.1  | 28.0  | 28.5  | 30.1  | 30.6  | 30.6  | 30.2  | 29.2  | 27.6  | 26.8  |
|                                                                        | 10%         | WB          | 22.7  | 22.7  | 22.7  | 23.7  | 24.7  | 25.7  | 25.8  | 26.0  | 25.9  | 25.4  | 24.3  | 23.7  |
|                                                                        |             | MCDB        | 25.7  | 26.1  | 26.6  | 27.5  | 28.1  | 29.7  | 30.2  | 30.3  | 29.8  | 28.8  | 27.1  | 26.5  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.6   | 6.0   | 6.1   | 5.8   | 5.7   | 6.0   | 6.0   | 6.0   | 6.1   | 5.1   | 4.9   | 5.2   |
|                                                                        |             | MCDBR       | 6.6   | 7.3   | 7.6   | 6.7   | 6.5   | 6.9   | 6.7   | 6.4   | 6.6   | 6.1   | 5.3   | 5.5   |
|                                                                        | 5% WB       | MCWBR       | 3.0   | 3.1   | 2.9   | 2.5   | 2.5   | 2.4   | 2.2   | 2.2   | 2.3   | 2.3   | 2.3   | 2.6   |
|                                                                        |             | MCWBR       | 6.2   | 6.0   | 6.6   | 5.9   | 5.7   | 6.4   | 6.2   | 6.2   | 6.2   | 5.7   | 4.7   | 5.2   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.364 | 0.373 | 0.388 | 0.409 | 0.434 | 0.469 | 0.474 | 0.450 | 0.421 | 0.413 | 0.389 | 0.386 |
|                                                                        |             | taud        | 2.571 | 2.541 | 2.489 | 2.421 | 2.397 | 2.324 | 2.297 | 2.393 | 2.489 | 2.499 | 2.520 | 2.519 |
|                                                                        | Ebn at Noon |             | 906   | 918   | 915   | 895   | 864   | 828   | 824   | 847   | 871   | 865   | 871   | 867   |
|                                                                        |             | Edh at Noon | 92    | 100   | 110   | 119   | 121   | 129   | 133   | 121   | 108   | 103   | 96    | 94    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 4.01  | 4.95  | 5.73  | 6.43  | 6.24  | 5.97  | 6.47  | 6.14  | 5.51  | 4.72  | 4.15  | 3.80  |
|                                                                        | RadStd      |             | 0.25  | 0.20  | 0.23  | 0.23  | 0.51  | 0.37  | 0.36  | 0.25  | 0.25  | 0.31  | 0.24  | 0.27  |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (4 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page